

FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10: TRIGONOMETRY I (3.1) FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Driving question: How can the angle a sunbeam makes with the ground allow us to calculate the height of any tree, building, or hill in Kenya without measuring it directly? Use what you learned in all ten lessons to answer the prompts below. Write in full sentences, show every step of any calculation, and include units. Draw labelled right-angled triangles wherever they help. Remember the Kisumu data: a student 1.6 m tall casts a 1.2 m shadow, the mango tree casts a 9.0 m shadow, and the sun angle measured with the clinometer is about 53° . Be honest about any rounding or measurement error in your work.

Part 1: Why does the angle tell us about the sides?

In the Kisumu school compound, the student and the mango tree both stand in the same sunlight. Draw and label the two right-angled triangles (one for the student and her shadow, one for the tree and its shadow). Explain why the ratio of opposite to adjacent is the same in both triangles, even though one triangle is much bigger. Then explain why this means that an angle can tell us about the ratios of the sides of a right-angled triangle.

Write your answer here:

Part 2: Sine, cosine and tangent, and how they are related

(a) State the definitions of $\sin \theta$, $\cos \theta$ and $\tan \theta$ for an acute angle θ in a right-angled triangle. (b) Starting from these definitions, show that $\tan \theta = \sin \theta \div \cos \theta$. (c) Using $\sin 53^\circ \approx 0.7986$ and $\cos 53^\circ \approx 0.6018$ from the tables, check this relationship for 53° and say how your answer compares with $\tan 53^\circ \approx 1.3270$.

Write your answer here:

Part 3: Finding the height of the mango tree

(a) Use the sun angle of 53° , the 9.0 m shadow and $\tan 53^\circ \approx 1.3270$ to calculate the height of the mango tree. Show which side is opposite and which is adjacent. (b) Find the height a second way, using the student's height and shadow and similar triangles. (c) Compare your two answers. Why are they slightly different, and what should we say about the tree's height? Explain why you did not need to climb the tree.

Write your answer here:

Part 4: Special angles and complementary angles

(a) Suppose the sun in Kisumu made an angle of exactly 60° with the ground and the mango tree's shadow was still 9.0 m. Using exact values only (no calculator), find the height of the tree. Do the same for an angle of exactly 45° . (b) In the 53° triangle, the other acute angle is 37° . Explain how $\sin 53^\circ$ and $\cos 37^\circ$ are related and why. Write the identity that you used.

Write your answer here:

Part 5: Answering the driving question for any height

(a) Suppose there is no shadow, for example a school flagpole or a Safaricom mast in Kericho. Describe how you could find its height using a clinometer and a tape measure. Name the angle you measure, the length you measure, and the ratio you use. **(b)** Compare your Lesson 1 hand-drawn ratio estimates with the calculator values. Explain why they might differ, and why it is important to report any difference honestly. **(c)** Now write a short answer to the driving question and to the two KICD questions: How do we apply trigonometric ratios in day-to-day activities? How are the trigonometric ratios of acute angles related to each other?

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of similar triangles and ratio ideas	Clearly explains that triangles with the same angle are similar, so the ratios are fixed by the angle alone, and links this to the Kisumu data.	States that the ratios stay the same for the same angle but gives only a partial reason about similarity.	Describes the ratios without explaining why they stay constant, or confuses the sides.
Correct calculations and exact values	Correctly finds the tree height (about 11.94 m and 12.0 m) and the special-angle heights ($9\sqrt{3}$ m and 9.0 m), showing every step with units and exact values.	Calculations are mostly correct with minor slips in steps, units or rounding.	Calculations are incomplete or incorrect, or the opposite and adjacent sides are mixed up.
Relationships among the ratios	Proves $\tan \theta = \sin \theta \div \cos \theta$ and explains $\sin \theta = \cos(90^\circ - \theta)$ using the swap of opposite and adjacent sides.	States both relationships correctly and checks them with numbers, but the explanation is brief.	States a relationship without a reason or with errors.
Application, reasoning and integrity	Describes a complete method for finding any height from an angle and a baseline, explains measurement error honestly, and answers the driving question clearly with Kenyan examples.	Describes a workable method and mentions error, with a general answer to the driving question.	Method is unclear or missing, and error and honesty are not discussed.

ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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